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Multiscale characterization of shale softening induced by water-based fluids

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ABSTRACT

Characterization of mechanical alterations of shale constituent phases is critical for an in-depth understanding of the underlying mechanisms of shale softening. In this study, a hydro-thermal reaction system is set up to mimic the interactions between shale and water-based fluids under the subsurface environment in shale formations. Using a coupled analysis of grid nanoindentation and in situ mineralogical identification, mechanical alterations of shale constituent mineral phases are revealed. Mechanical degradation of carbonate and clay phases is 10 times greater than quartz, pyrite and organic phases. The KCl additive greatly mitigates mechanical degradation of the clay phase. The high temperature and pressure results in a mechanical degradation of carbonate minerals as much as three times of that occurs at room temperature and atmospheric pressure. Multiscale mechanical models, which are established based on Mori-Tanaka (MT) and self-consistent (SC) schemes, predict more accurate elastic softening of shale composite than the microindentation experiments, due to the microcracks generated in the experiments. Based on the calculation of the multiscale mechanical model, under the subsurface environment of shale formations (e.g. 80 °C and 8 MPa), the carbonate dissolution leads to a reduction in Young's modulus of shale composite by about 30%, while the degradation of clay minerals only causes a reduction by up to 9%.

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1. Introduction

Shale gas exploration has attracted significant attention due to the growing challenges in meeting global energy demand (Du et al., 2018). To achieve economically viable gas production of shale gas, hydraulic fracturing fluids are loaded with chemical additives (e.g. KCl) to create extensive fracture networks in tight shale formations (Chen et al., 2021). However, fluid interactions that take place in these fractures can cause significant mechanical alterations shale formations, which can result in a series of negative engineering consequences, such as the reduction in gas recovery (Song et al., 2021), wellbore instability (Ibrahim, 2021), and reservoir subsidence (Shad et al., 2023). To tackle the engineering challenges, an in-depth and comprehensive understanding of shale softening is required.

The majority of existing studies have investigated the mechanical alterations of shale using conventional macroscopic experiments, such as triaxial shear and uniaxial compression tests (Li et al., 2020). These tests use core specimens that require expensive, time-consuming sampling process (Graham et al., 2021). Furthermore, shale is a composite material composed of grain minerals and highly consolidated aggregates of clay particles and organic matter (Du et al., 2022). Conventional macroscopic tests cannot reflect the heterogeneous and multiscale characteristics of shale's mechanical properties. Fig. 1 shows the conceptional multiscale structure of shale composite. Conventional laboratory experiments are focused on the macroscopic shale, i.e. Level IV. Level III is the scale of the elementary unit of a shale composite. Levels I and II represent the scales of constituent mineral phases. Therefore,

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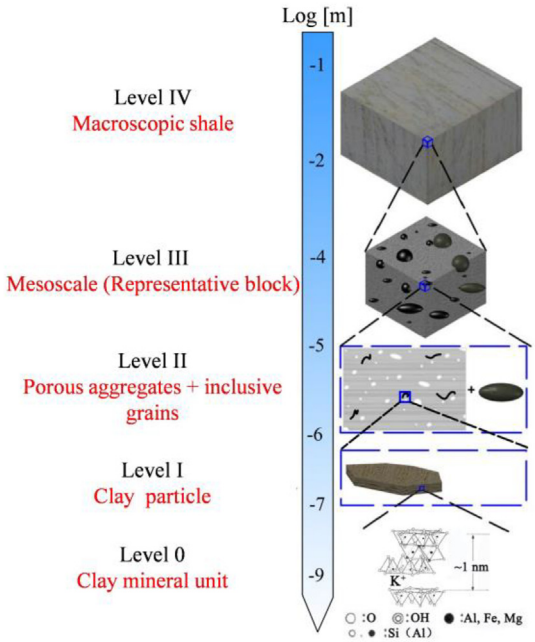


Fig. 1. Conceptional multiscale structure of shale (Du et al., 2023).

mechanical alterations of shale constituent phases are fundamental knowledge for the multiscale characterization of shale degradation, and can provide an in-depth understanding of the underlying softening mechanisms.

Over the past decades, the nanoindentation technique was developed to assess the mechanical properties of composite materials at the nanoscale (Oliver and Pharr, 1992). Recently, the nanoindentation methods have been employed to investigate the shale softening behavior at nano-to micro-scales. Li et al. (2023) analyzed the mechanical degradation of shale using nanoindentation tests. Mechanical alterations of three phases of shale (i.e. soft, intermediate, and stiff phases) were measured by fitting the probability density function (PDF) of the experimental data with a number of theoretical PDFs. Liu et al. (2021b) and Cheng et al. (2024) probed the nanomechanical properties of shale after the treatment of water and CO_2. The Gaussian mixture model (GMM) combined with *k*-means or expectation maximization (EM) algorithm was used to fit the large-volume nanoindentation data, and the mechanical degradation of three clusters were discussed. However, in these studies, mineral phases of shale were roughly divided into three groups, i.e. high, medium, and low strength minerals, though statistical analyses of the nanomechanical data. Mechanical alterations of the individual mineral phases induced by the shale-fluids interactions were still unclear. Recently, Liu et al. (2024) assessed the softening behavior of constituent mineral phases of Marcellus shale by combining nanoindentation measurements and SEM (scanning electron microscopy)/BSE (backscatter electron imaging)-EDS (energy-dispersive X-ray spectroscopy). They matched each indentation measurement with its mineralogy visually based on the residual indentation footprint observed from the SEM/BSE imaging. However, the indentation data reflect the mechanical response of the rock within the elastoplastic deformed zone, which is typically 10 times the indentation depth (Durst et al., 2005) and is much greater than the residual indentation footprint. It is inefficient to conduct visual matching of nanomechanical results with mineralogical information, especially for statistical analysis involving large nanoindentation datasets (which aim to reduce the errors caused by experimental noise). Therefore, a more accurate and effective analysis is required to characterize the softening behavior of shale constituent phases.

Among the homogenization methods of composite materials, Mori-Tanaka (MT) and self-consistent (SC) schemes are the most representative ones (Wang et al., 2023), which have been widely used in the homogenization of geotechnical materials properties, e.g. rock and concrete (Zanjani Zadeh and Bobko, 2014; Cao et al., 2021, 2024; Hu et al., 2024). Zhao et al. (2019) predicted the macroscopic properties of Triassic Yanchang shale using MT and SC methods, and both methods exhibited a good agreement with the microindentation results. Li et al. (2019) upscaled the Young's modulus of Bakken shale using MT model, and found a reasonable agreement with core plug measurements. The MT scheme works effectively for composites consisting of continuous matrix and discontinuous inclusive phases (Punera and Kant, 2019), whereas the SC scheme exhibits good predictions for disordered polycrystalline materials (Mercier and Molinari, 2009). Therefore, multiscale mechanical models are established in this study based on both MT and SC schemes (depending on the morphology of the composite material, i.e. inclusion-matrix morphology or polycrystalline morphology) to analyze the upscaled softening behavior of shale composite.

In this paper, water-based fluids are used to interact with Longmaxi shale samples at different temperatures and pressures for 7 d, using a hydro-thermal reaction system. Mechanical alterations of shale constituent phases are then revealed via an auto-matching of nanomechanics with the local mineralogy. Subsequently, the degradation of elastic modulus of bulk shale is evaluated using a multiscale mechanical model, and compared with the experimental results obtained from the statistical microindentation measurements. Finally, relative contributions of clay and carbonate phases in mechanical degradation of shale composite are further discussed.

2. Materials and methods

2.1. Sample preparation

The shale samples are taken from the Silurian Longmaxi Formation in Sichuan Basin, China. The shale samples are examined under the optical microscope, and no visible bedding planes are observed. This may be attributed to the mineralogy of the Longmaxi shale samples, which presents a low content of clay minerals (<30%) and high content of carbonate and quartz minerals (>60%) (see Section 2.2 for details of the mineralogy of the shale samples). Before the nanoindentation measurements, the tested samples are first trimmed into small cuboids and then mechanically polished. The surface roughness is assessed by atomic force microscope (AFM). The measured surface roughness after polishing is 16 nm.

In the subsurface environment, shale interacts with the injected fracturing fluids at elevated temperatures and pressure. Therefore, in this study, a hydro-thermal reaction system is used to treat the shale samples. A heating belt and a Nitrogen tank are used in the reaction system to provide the target reaction temperature and pressure. In this study, the short-term softening behavior of shale after exposure to water-based fluids is focused; therefore, the intact shale samples are soaked in water-based fluids for a 7-d duration. In practice, 2% KCl is a common additive of fracturing fluids to maintain wellbore stability (Rasool and Ahmad, 2024). Therefore, DI water and 2% KCl solution are used in this study to analyze the effect of KCl on the shale softening behavior. For further details of the hydro-thermal reaction system, the reader is referred to Du et al. (2024). The treatments used in the hydro-thermal reaction system are summarized in Table 1.

Table 1
The treatments used in the hydro-thermal reaction system.

Sample	Soaked solution	Temperature (°C)	Pressure (MPa)	Duration (d)
S1		20	0.1	7
S1*	De-ionized water	20	0.1	7
S2		20	0.1	7
S2*	2% KCl brine	20	0.1	7
S3		20	0.1	7
S3*	De-ionized water	80	8	7
S4		20	0.1	7
S4*	2% KCl brine	80	8	7

Note: The intact and treated samples are named as S#, and S#*, respectively.

2.2. In situ microstructural and mineralogical characterization

The in situ microstructural and mineralogical alterations of shale specimens induced by the treatments are characterized using a microstructural and mineralogical identification technique (see Fig. 2). First, before and after the treatments, the same region (with an area of 240 μm × 300 μm), where the grid nanoindentation is located, is positioned under SEM imaging, followed by the BSE imaging and EDS mapping. Second, the voids in the region of interest (ROI) are extracted from the BSE images using a threshold algorithm of grayscale (Singh et al., 2020). Third, the quantitative elemental information is extracted from each pixel, and based on the chemical composition of shale constituent phases, a pixel-by-pixel identification of in situ mineralogy of the ROI is performed through Matlab coding. Finally, a 'digital mineral map' comprising the spatial information and area fractions of shale mineral phases is generated.

Fig. 3 shows the mineral map of sample S1. The primary minerals contained in the ROI are carbonate minerals (i.e. calcite and dolomite: ~40%), clay (~26%), quartz (~22%) and pyrite (~2%). Due to the limitation of the resolution of EDS scan, multiple mineral phases can be measured in an individual pixel (0.3 μm × 0.3 μm), such as a mixture of clay and calcite was identified in the mineral map of S1 (shown as CL&CA).

2.3. Nanomechanics and local mineralogy coupling

A square grid of 21 × 21 nanoindentation measurements is carried out on each polished sample, with a spacing of 10 μm between indentation points. Mechanical parameters, i.e. Young's modulus, E, and hardness, H, are calculated from the indentation force versus penetration depth curves measured in the grid nanoindentation (Du et al., 2023). Due to the clay content in the shale samples, time-dependent behavior will occur during the nanoindentation measurements (Sone and Zoback, 2014; Bao et al., 2021). To diminish the time-dependent plastic effects, a 180-s holding phase is applied at the peak load to allow the sample to creep and Young's modulus is measured based on the initial unloading slope at the peak load (Oliver and Pharr, 1992, 2004; Feng and Ngan, 2002). To assess the mechanical properties of shale constituent phases, the indentation depth should be small enough to avoid multiple mineralogical phases involved in each indentation volume (i.e. elastoplastic stressed zone formed at each indenter location). However, the indentation depth is restricted by the surface roughness achieved, i.e. $h > 3R_q$ (Donnelly et al., 2006). Based on the surface roughness measured in this study, a maximum depth of 100–400 nm (depending on the stiffness of the indented phase) was selected.

Mechanical properties of shale constituent phases are determined by linking the measured nanomechanical information to the local mineralogy. As shown in Fig. 4, the coordinates of the nanoindentation grid in the ROI are extracted from the BSE images based

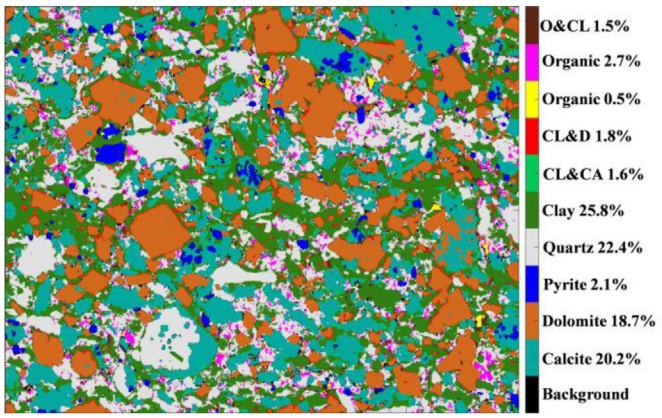


Fig. 3. In situ mineralogical identification of sample S1. CL, Q, CA, D, and O are abbreviations of clay, quartz, calcite, dolomite, and organic, respectively.

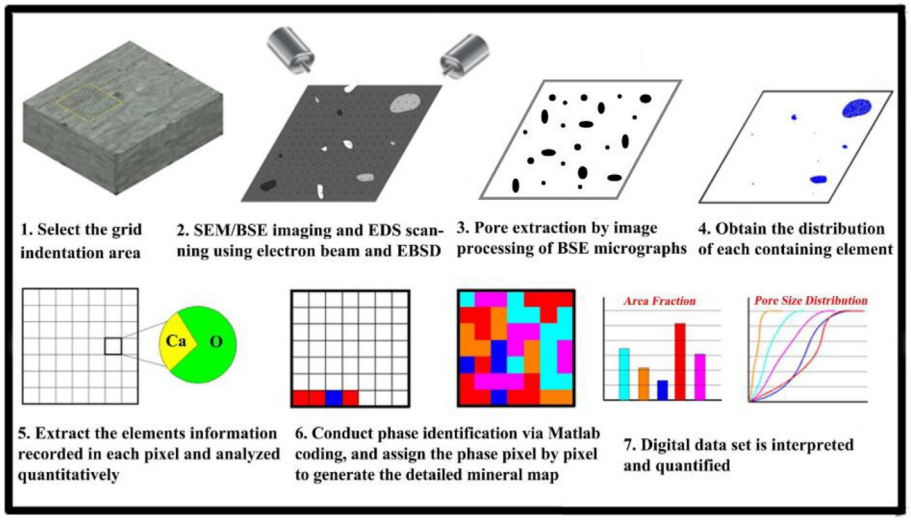


Fig. 2. Procedures for in situ microstructural and mineralogical characterization.

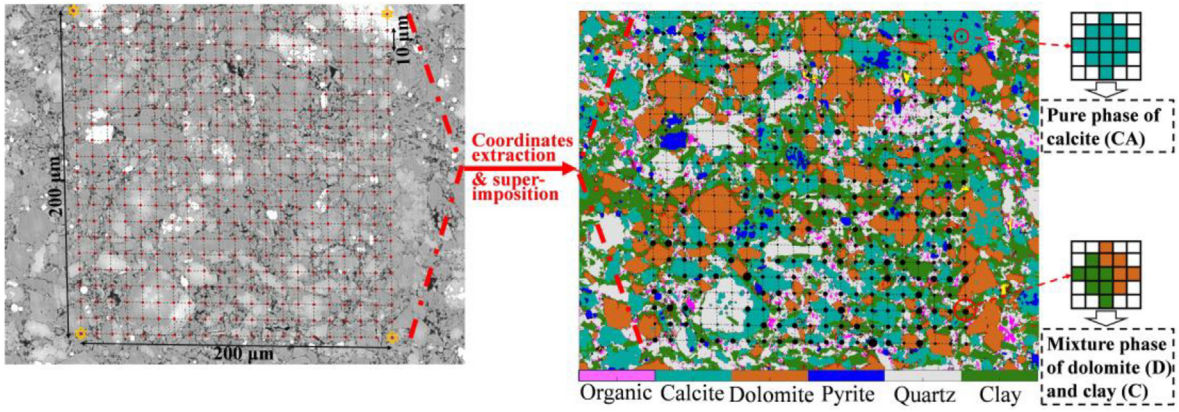


Fig. 4. Procedures for linking the measured nanomechanical information with the local mineralogy.

on the residual indentation of four grid corners and the test protocol (measurements are carried out sequentially in horizontal and vertical directions with a spacing of 10 μm between each two measurements). Then the nanoindentation grid is superimposed on the digital mineral map based on their coordinates. The area of black dots in the mineral map in Fig. 4 reflects the size of the indentation interaction volume. The mineral phases involved in each indentation interaction volume are determined based on a conservative rule of thumb: a pure mineral phase is identified if more than 90% pixels display the same mineralogy (e.g. calcite phase; in Fig. 4). Otherwise, the measurement is corresponded to a mixture phase if multiple phases are present with a proportion >10% (e.g. mixture phase of dolomite and clay; in Fig. 4).

2.4. Flow chart of experimental analyses on mechanical degradation of shale constituent phases

Fig. 5 shows the flow chart of the experimental analyses in this study, which mainly includes 6 steps.

(I) The intact shale samples are polished and measured by the grid nanoindentation.
(II) The in situ mineralogy, morphology, and porosity in the ROI of the intact samples are determined before the treatments.
(III) Water-based fluids are used to interact with shale at different temperatures and pressures for 7 d.
(IV) After each treatment, alterations of surface mineralogy, morphology, and porosity in the ROI are measured.
(V) The surface roughness after the treatments will be increased due to the precipitations from shale-fluids reactions. Therefore, the treated samples are repolished to eliminate the influence of the increased surface roughness on the nanoindentation measurements, and then grid nanoindentation tests are conducted again on the repolished samples. It is noted that, based on the color change between the intact zone and the zone with water infiltration, the depth of repolishing has been confirmed by SEM/BSE imaging (hundreds of nm to a few μm), which is two-orders smaller than that of the shale-fluid interaction zone (hundreds of μm), to ensure the mechanical alterations caused by the treatments of water-based fluids are measured by the nanoindentation tests.
(VI) The mechanical properties of mineral phases before and after each treatment are extracted by the coupled analysis of nanomechanics and local mineralogy (see Section 2.3 for the details of the coupled analysis), and then the measured mechanical properties of each phase are further compared to analyze the mechanical alterations of shale constituent phases after different treatments. The alterations of surface mineralogy, morphology, and porosity in the same ROI are compared to analyze the quantitative changes in mineral fraction and void ratio.

2.5. Upscaling mechanical models

Multiscale mechanical models are established in this study based on both MT (inclusion-matrix morphology) and SC (polycrystalline morphology) schemes to analyze the upscaled softening behavior of shale composite. The Young's moduli of the intact and degraded shale composite are estimated based on the elastic stiffnesses and area fractions of constituent phases, and the void ratio of each specimen before and after the treatments. For linear elastic behavior of the mineral phases, linear micromechanics theory (Li et al., 2017; Mashhadian et al., 2018) provides the following estimate of the homogenized stiffness tensor of the solid-pore composite:

$$\mathbb{C}^{\mathrm{hom}} = \langle \mathbb{C} : \mathbb{A} \rangle_{\Omega} = \sum_{i=1}^{N} \varphi_i \mathbb{C}_i : \mathbb{A}_i \tag{1}$$

where Ω is the domain of the representative elementary volume (rev) centered at point $\underline{z}$; the angled brackets define a volume average over the rev domain Ω; $\mathbb{C}$ denotes the stiffness tensor, and $\mathbb{A}$ is the fourth-order strain localization tensor which linking the macroscopic strain and the microscopic strain; φ_i, $\mathbb{C}_i$, and $\mathbb{A}_i$ are the volume fraction, stiffness tensor, and strain localization tensor of the i-th phase, respectively; and N is the number of the constituent phases. Considering the Eshelby inclusion theory (Eshelby, 1957), the strain localization tensor is obtained as follows:

$$\mathbb{C}^{\mathrm{hom}} = [I + \mathbb{H}_0 : (\mathbb{C}_i - \mathbb{C}_0)]^{-1} : \left\{ \sum_{i=1}^{N} \varphi_i [I + \mathbb{H}_0 : (\mathbb{C}_i - \mathbb{C}_0)]^{-1} \right\}^{-1} \tag{2}$$

where $\mathbb{C}_0$ is the stiffness tensor of the embedding medium; I is the unit tensor; and $\mathbb{H}_0$ is the Hill tensor, given by

$$\mathbb{H}_0 = \frac{1}{3K_0 + 4G_0} \mathbb{J} + \frac{3}{5G_0} \frac{K_0 + 2G_0}{3K_0 + 4G_0} \mathbb{K} \tag{3}$$

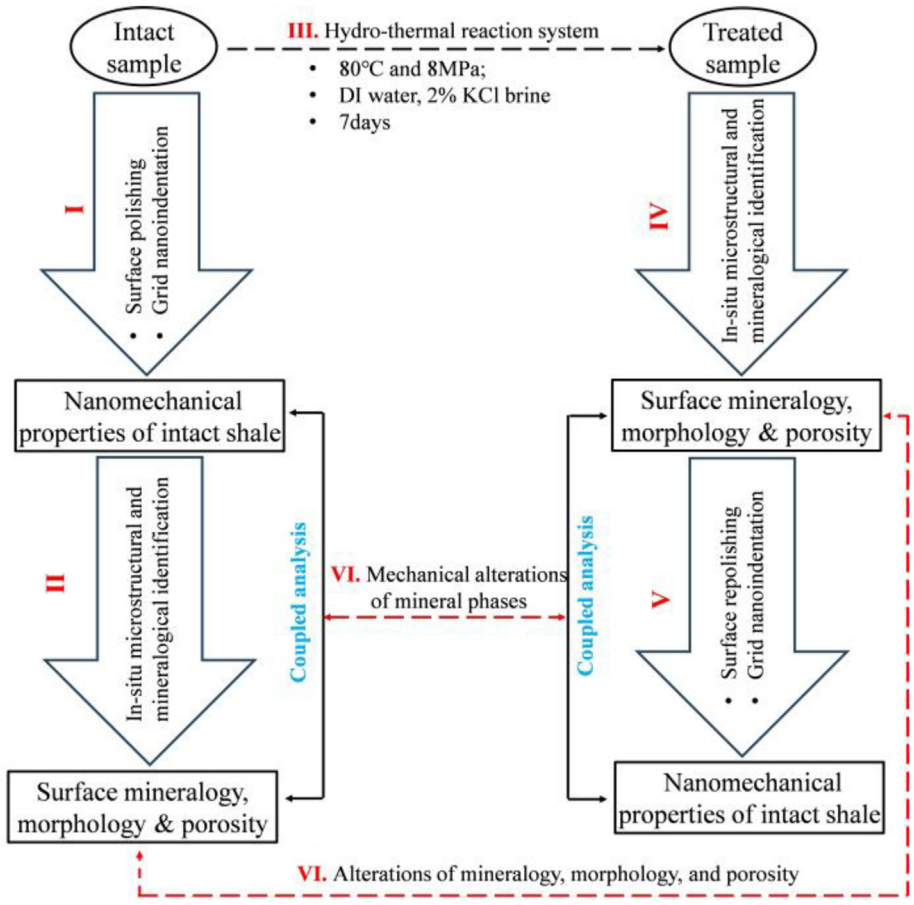


Fig. 5. Flow chart of the experimental analyses on the mechanical degradation of shale constituent phases.

where K_0 and G_0 are the bulk modulus and shear modulus of the embedding medium, respectively; and $\mathbb{J}$ and $\mathbb{K}$ are the volumetric part and the deviatoric part of the fourth-order unit tensor, respectively.

Substituting Eqs. (2) and (3) into Eq. (1), the homogenized bulk modulus and shear modulus can be obtained as follows:

$$K_{\text{hom}}=\sum_i \varphi_i K_i\left[1+\alpha_0\left(\frac{K_i}{K_0}-1\right)\right]^{-1}\left\{\sum_i \varphi_i\left[1+\alpha_0\left(\frac{K_i}{K_0}-1\right)\right]^{-1}\right\}^{-1} \tag{4}$$

$$G_{\text{hom}}=\sum_i \varphi_i G_i\left[1+\beta_0\left(\frac{G_i}{G_0}-1\right)\right]^{-1}\left\{\sum_i \varphi_i\left[1+\beta_0\left(\frac{G_i}{G_0}-1\right)\right]^{-1}\right\}^{-1} \tag{5}$$

where $\alpha_0=\frac{3K_0}{3K_0+4G_0}$, $\beta_0=\frac{6(K_0+2G_0)}{5(3K_0+4G_0)}$.

For a composite whose morphology comprises inclusions embedded within a matrix, the mechanical model should conform to the MT scheme (Mori and Tanaka, 1973), such that K_0 and G_0 are given by the bulk modulus and shear modulus of the matrix phase in the composite material. In contrast, for a composite with a granular morphology, where there is no dominant matrix phase, the mechanical model should follow the SC (or polycrystal) scheme (Hershey, 1954) and the constituent phases are considered being embedded into the homogenized composite itself. Therefore, K_0 and G_0 in Eqs. (4) and (5) are defined as the bulk modulus and shear modulus of the homogenized composite, i.e. $K_0=K_{\text{hom}}$, $G_0=G_{\text{hom}}$. Then the homogenized moduli can be obtained by solving Eqs. (4) and (5) numerically. Prior studies showed that the value of Poisson's ratio has little influence on the calculation of the homogenized Young's modulus (Randall et al., 2009). In this study, Poisson's ratio is determined as 0.2 based on the typical ranges of Longmaxi shale reported in the literature (Tian et al., 2021).

It is noted that the upscale model for Levels I and II (refer to Fig. 1) can be established based on the MT scheme (Eqs. (4) and (5)) by considering the porous clay aggregates as a two-phase composite consisting of clay particles and voids.

3. Results

The mechanical alterations of shale constituent phases induced by water-based fluids at different temperatures and pressures are characterized. Furthermore, an upscaling analysis is carried out based on MT and SC schemes to evaluate the elastic modulus alteration of bulk shale.

3.1. Mechanical alterations of shale constituent phases

Mechanical properties of each shale constituent phase before and after the treatments of water-based fluids at different temperatures and pressures are compared. Figs. 6 and 7 show alterations of E and H of shale constituent phases (i.e. five primary mineral phases: calcite, dolomite, clay, quartz and pyrite, and organic matter) after different treatments. The results are

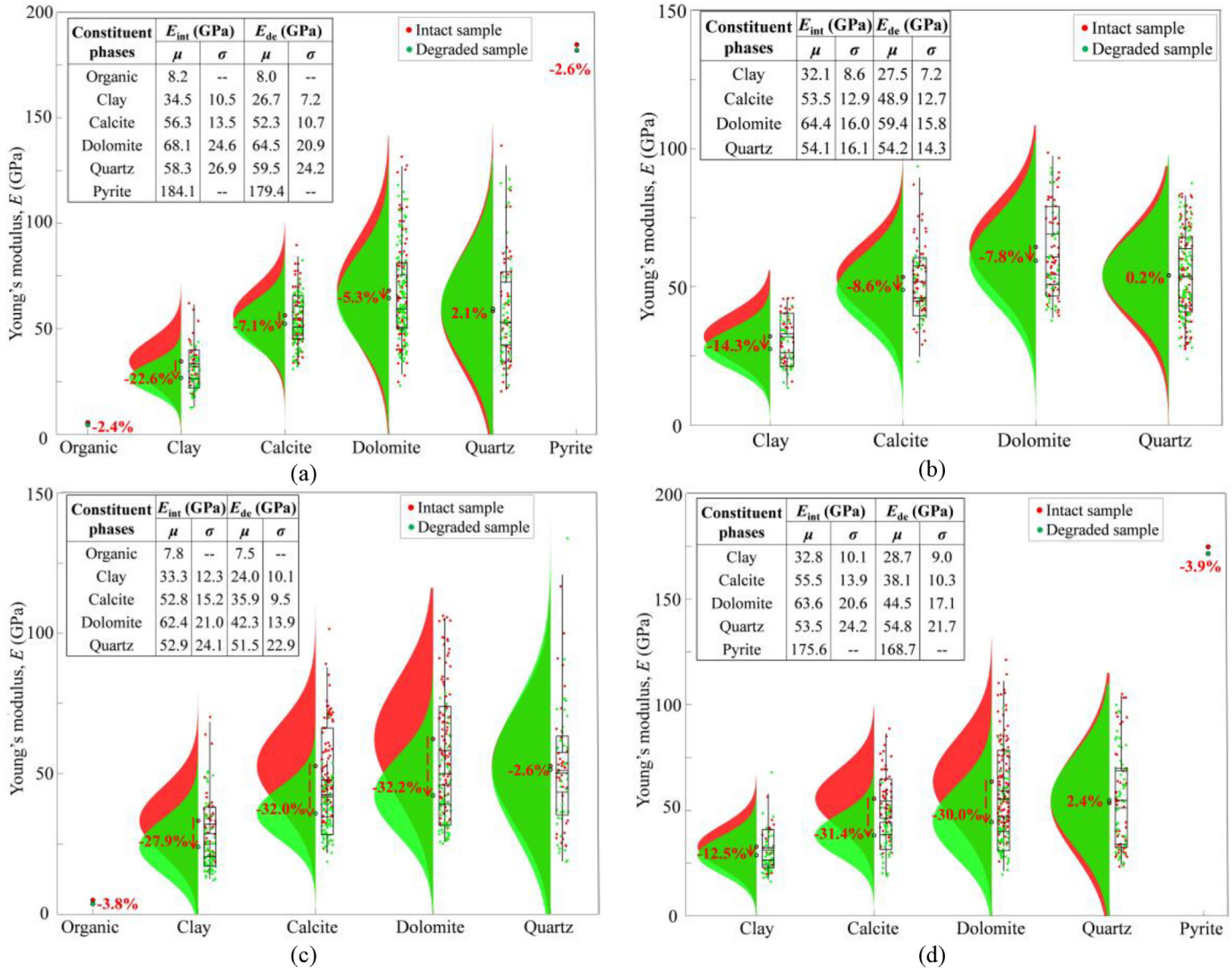


Fig. 6. Alterations of Young's modulus of shale constituent phases after different treatments: (a) DI water at 20 °C and 0.1 MPa, (b) 2% KCl brine at 20 °C and 0.1 MPa, (c) DI water at 80 °C and 8 MPa, and (d) 2% KCl brine at 80 °C and 8 MPa. The black circles at the bottom of Gaussian distributions show their statistical mean, μ. E_{int} and E_{de} represent the Young's moduli of intact and degraded phases, respectively.

presented in a raincloud format, which combines conventional boxplots with statistical interpretations of Gaussian distributions (Allen et al., 2019). Due to the small content of pyrite and organic matter in the studied shale samples, limited data were obtained and the arithmetical average is used to assess their E and H. The results show that significant mechanical degradations occur in clay and carbonate phases (a maximum reduction of 32% and 35% in Young's modulus and hardness, respectively), whereas only minor mechanical alterations (less than 3%) are observed for organic, quartz, and pyrite phases.

Compared with the treatment of DI water, the addition of 2% KCl in the brine solution greatly mitigates the mechanical degradation of the clay mineral. The 7-d exposure to DI water induces a reduction of ~25% and ~27% in E and H of the clay phase, whereas the 2% KCl brine only causes a reduction of ~13% and ~10% in E and H, respectively. Based on the SEM/BSE imaging, inner-cracks of the clay phase due to the clay swelling are observed after the treatment of DI water for 7 d (see Fig. 8), which may be the main reason for the mechanical degradation of clay phase (reduction of 12.5%–27.9% and 9.3%–29.6% for E and H, respectively). The clay minerals will be degraded due to the clay swelling, which typically occurs via two schemes, i.e. crystalline swelling and osmotic swelling (Rao et al., 2013). Existing research showed that the temperature and pressure primarily affect the barrier to swelling and the chemical potential (i.e. electrolyte concentration) determines the layer spacing (Zhang et al., 1993; Smith et al., 2006). The temperature and pressure conditions adopted in this study are beyond the barrier point and thus show little to no effect on the clay swelling, whereas the 2% KCl brine significantly mitigates the swelling behavior of clay phase. An increase of the solution concentration prevents the infiltration of water into the porous clay matrix, which thus suppresses the osmotic swelling of the clay phase. The elevated temperature and pressure exhibit negligible effects on the mechanical degradation of clay phase; however, they play critical roles in the mechanical degradation of the carbonate phases (i.e. calcite and dolomite). The mechanical degradation of carbonate minerals at 80 °C and 8 MPa is increased by more than 3 times compared to that at 20 °C and 0.1 MPa. The reduction of E and H of carbonate minerals ranges from 5% to 9% after the exposure to water-based fluids at 20 °C and 0.1 MPa, whereas a mechanical degradation of ~30% is

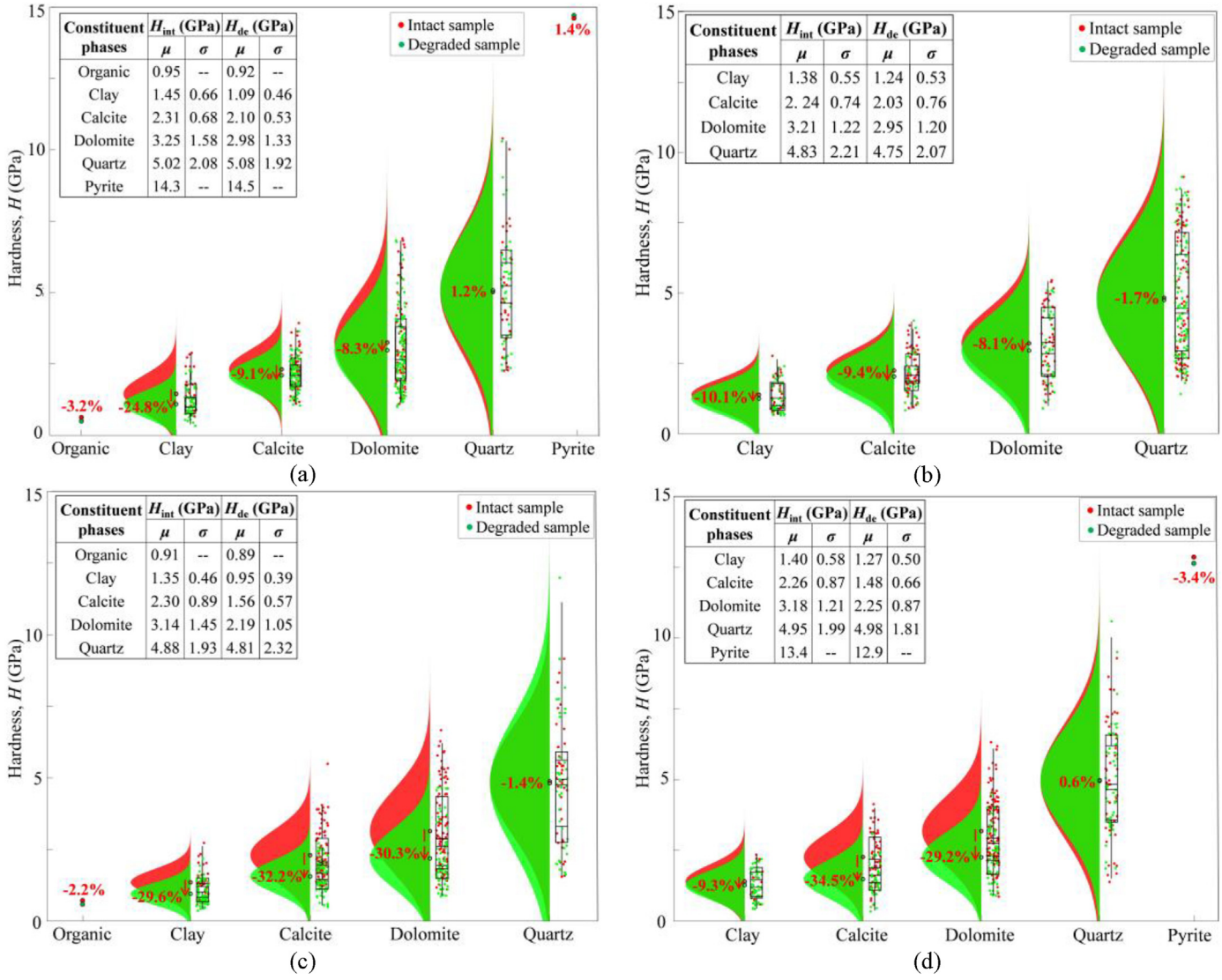


Fig. 7. Alterations of hardness of shale constituent phases after different treatments: (a) DI water at 20 °C and 0.1 MPa, (b) 2% KCl brine at 20 °C and 0.1 MPa, (c) DI water at 80 °C and 8 MPa, and (d) 2% KCl brine at 80 °C and 8 MPa. The black circles at the bottom of Gaussian distributions show their statistical mean, μ. H_{int} and H_{de} represent the hardnesses of intact and degraded phases, respectively.

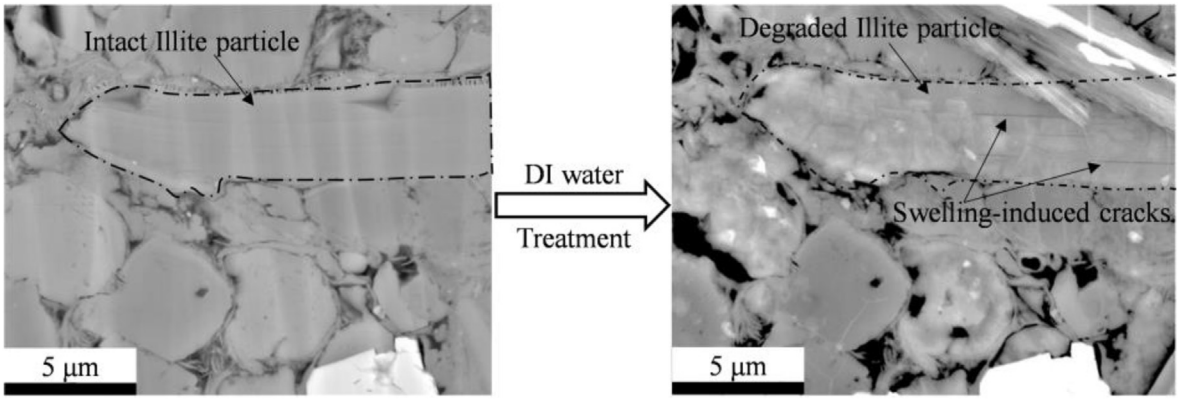


Fig. 8. Inner-cracks of the clay phase observed by BSE imaging after the treatment of DI water for 7 d.

observed at 80 °C and 8 MPa. Microstructural and mineralogical alterations induced by the physicochemical interactions between shale mineral phases and injected fluids govern the mechanical degradation of shale. The carbonate minerals calcite ($CaCO_3$) and dolomite ($CaMg(CO_3)_2$) are partially dissolved in the water-based solutions, and their dissolution rates increase with elevated reaction temperature and pressure (Kirstein et al., 2016). Fig. 9 shows the surface erosion caused by DI water at 8 MPa and 80 °C for 7 d. Based on the identification of in situ morphology and mineralogy, significant dissolution of surficial carbonate minerals is observed, while the pyrite and quartz phases remain intact. The findings indicate that, as the pressurized fracturing fluids are injected into the subsurface environment, the mechanical degradation of the clay content can be suppressed by the KCl additive, but the degradation of carbonate minerals may be greatly exacerbated due to the high temperature and pressure conditions in the shale formations.

3.2. Mechanical degradation of shale composite estimated by the upscaling models

Fig. 10 shows the results of void extraction of specimens S1 and S3 before and after the degradation. The voids are highlighted in blue and the surface void ratio corresponds to the area fraction of voids in each ROI. Furthermore, the area fractions of constituent phases in degraded specimens are estimated based on those measured from the intact specimens (using the in situ mineralogy identification, e.g. Fig. 3) and the increments of surface voids ratio. Based on the BSE observations, the generation of voids after degradation is mainly caused by the dissolution of carbonate minerals. Therefore, the area fractions of carbonate minerals in degraded specimens are estimated by reducing the increment of voids ratio from the values for the intact specimen, and for clay, quartz, pyrite and organic matter, the area fractions are assumed the same as those of intact specimens. The results show that the carbonate dissolution induces both degradation of carbonate phases and void ratio increase. Significant mechanical degradations occur in clay and carbonate phases (a maximum reduction of 32% and 35% in Young's modulus, respectively), whereas only minor mechanical alterations (less than 3%) are observed for organic, quartz, and pyrite phases. Mechanical degradation of carbonate and clay phases is 10 times greater than that of quartz, pyrite and organic phases, which indicates clay and carbonate are the causative minerals of the mechanical softening of the Longmaxi shale samples. The elastic stiffnesses and area fractions of constituent phases, and the void ratios for intact and degraded specimens are summarized in Table 2.

Fig. 11 shows the comparison between the calculated and measured E of intact and degraded shale composites. The measured E was obtained from the same shale specimens using grid microindentation tests. For complete details of the measured E of intact and degraded shale composite, the reader is referred to Du et al. (2024). Compared with the MT scheme, the SC scheme provides a greater homogenized E (see Fig. 11). This is because the elastic stiffness of the matrix phase used in the MT scheme (i.e. clay phase) is lower than that assumed in the SC scheme (i.e. the shale composite itself). Shale is typically recognized as a composite made of inclusive minerals, e.g. calcite, dolomite, quartz and pyrite, embedded into the clay matrix. However, based on the results from BSE imaging and mineralogy identification (see Figs. 3 and 10), no dominant matrix phase was observed in the Longmaxi shale, and hence its morphology is better described as a granular pattern. Therefore, the upscaling model built based on SC scheme is recommended to define the multiscale elastic stiffness of the Longmaxi shale used in this study.

Based on both MT and SC schemes, the calculated Young's moduli of intact shale composites are much larger than the experimental values; in contrast, the calculated and measured Young's moduli of degraded shale composites are close to each other. This may be because of microcracks generated during the microindentation tests. Prior studies showed that mechanical degradation can lead to a reduction in rock brittleness, which thus decreases the fracability, i.e. the capability of being fractured (Kang et al., 2020). Fig. 12 shows the residual microindentation footprints of specimens S1 and S1*. Massive microcracks are generated as the indenter penetrates the surface of the intact samples, whereas there are no obvious microcracks in the degraded samples (Fig. 12b). The generation of microcracks during the microindentation experiments results in an underestimation of the shale elastic modulus. Therefore, the calculated Young's moduli are greater than those measured on intact samples, but exhibit good agreement with those measured on the degraded samples (see Fig. 11).

The MT and SC models predict similar degradation in the homogenized E of shale composites after each treatment. The increased temperature and pressure result in degradations as much as twice those induced at 20 °C and 0.1 MPa. The KCl additive exhibits a mitigative effect on the mechanical degradation of shale composite. The reduction of homogenized E is changed from 17% to 13% at 20 °C and 0.1 MPa, and from 37% to 29% at 80 °C and 8 MPa, after adding 2% KCl into the DI water. Compared with the results calculated from the upscaling models, smaller decreases in E were measured from the grid microindentation measurements. Mechanical degradations of shale composite may be underestimated by the experimental

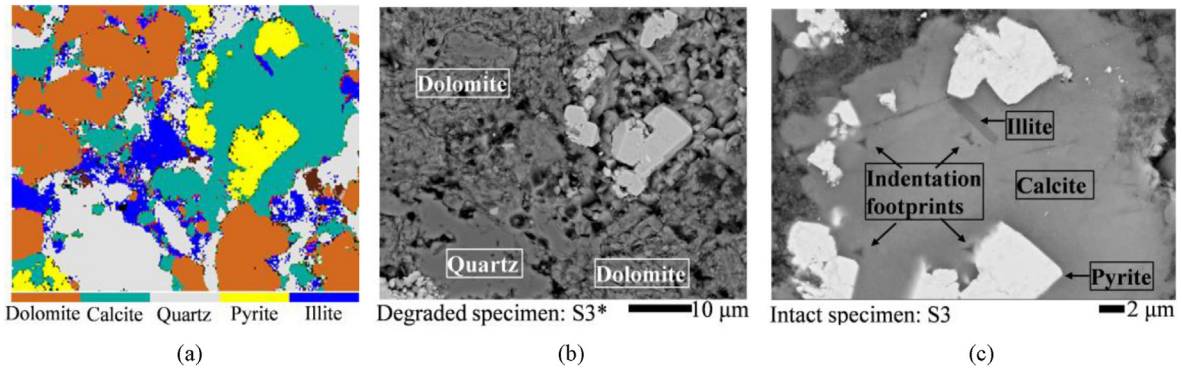


Fig. 9. Surface erosion caused by DI water at 8 MPa and 80 °C for 7 d: (a) In situ identification of in situ mineralogy in the ROI, (b) BSE micrograph of the ROI after degradation, and (c) BSE micrograph of the calcite crystal circled in (b) before degradation.

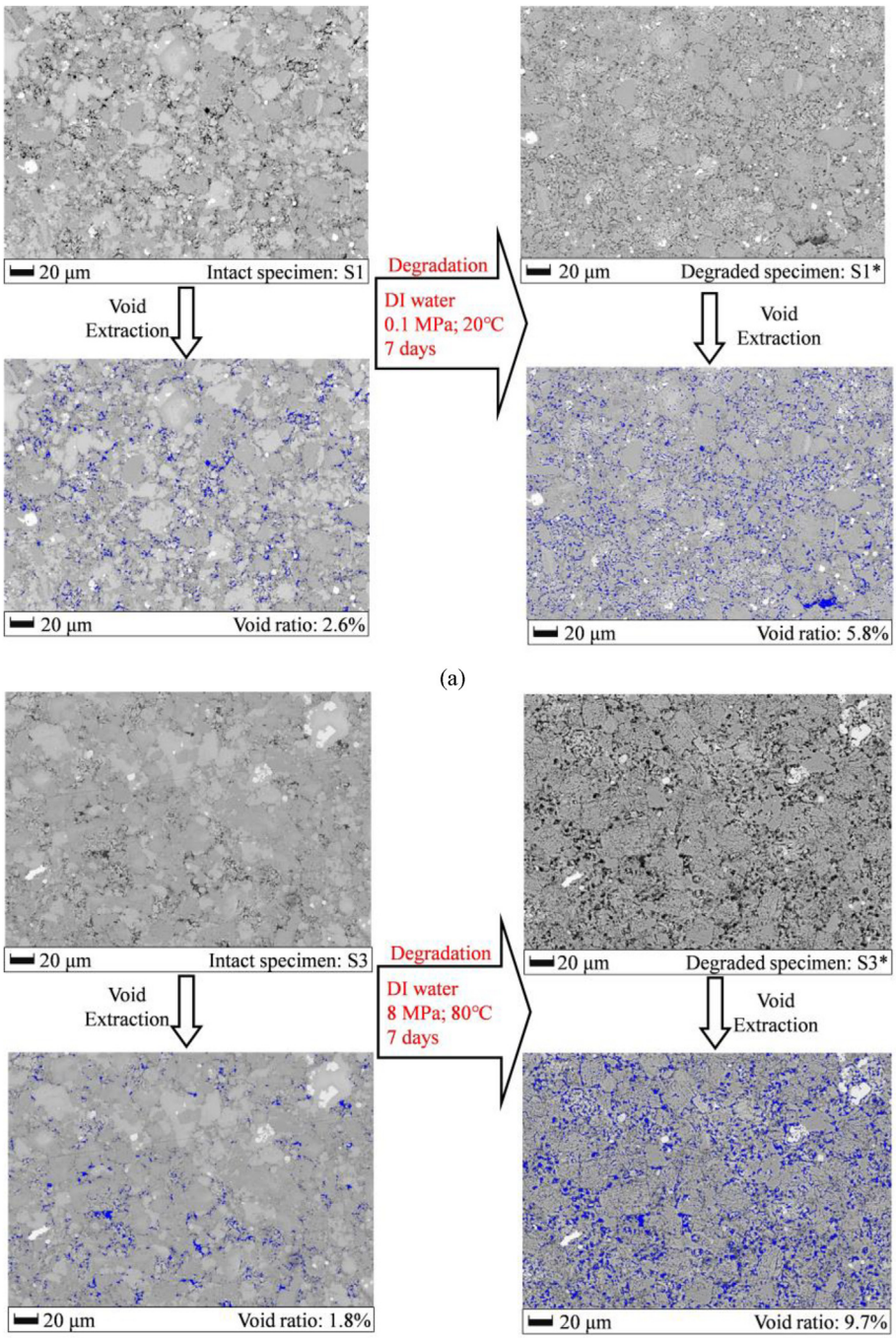


Fig. 10. Void extraction of specimens (a) S1 and (b) S3 before and after the degradation. The voids are highlighted with blue color.

measurements due to the massive microcracks generated when testing the intact shale samples. Compared with previous studies in the literature, the mineralogy of shale exhibits a critical impact on the shale softening after the treatment of water-based fluids. Liu et al. (2021b) studied the mechanical softening of Marcellus shale using nanoindentation and reported a reduction of 0.6%–15% in elastic modulus and a reduction of 8.6%–17.8% in hardness after 20-d water treatment. Unlike the Longmaxi shale samples used in this study, the Marcellus shale contains 40% illite and 40% silicate minerals (quartz and albite), but a minor content of carbonate mineral (2.8% ankerite). Akrad et al. (2011) and Corapcioglu et al. (2014) reported a reduction of ~70% and ~80% in Young's modulus of Eagle shale and Niobrara shale, respectively, after soaking the shale samples in the water-based fluids at 300 °F and 180 °F. Both two shale samples are abundant in carbonate minerals (>70% calcite). Moreover, the Ph of fluids also exhibit influences on the softening behavior of shale. Huang et al. (2024) compared the softening behavior of shale after soaking in brine and supercritical CO_2 + brine. The elastic modulus decreased by 29.86% and 44.99%, and the hardness decreased by 25.44% and 65.2% for the two soaking fluids, respectively. Results of the measured and calculated Young's modulus degradation of shale composites are summarized in Table 3.

Table 2
Results of the elastic stiffnesses and area fractions of constituent phases, and the void ratios for intact and degraded specimens.

Sample	Void		Organic matter		Clay		Calcite		Dolomite		Quartz		Pyrite	
	Elastic stiffness (GPa)	Area fraction (%)	Elastic stiffness (GPa)	Area fraction (%)	Elastic stiffness (GPa)	Area fraction (%)	Elastic stiffness (GPa)	Area fraction (%)	Elastic stiffness (GPa)	Area fraction (%)	Elastic stiffness (GPa)	Area fraction (%)	Elastic stiffness (GPa)	Area fraction (%)
S1	0	2.6	8.2	0.5	34.5	25.8	56.3	20.2	68.1	18.7	58.3	22.4	184.1	2.1
S1*	0	5.8	8	0.5	26.7	25.8	52.3	18.6	64.5	17.1	59.5	22.4	179.4	2.1
S2	0	3.2	(8)	2.6	32.1	23.1	53.5	19.4	64.4	22.4	54.1	23.9	(179.9)	2.3
S2*	0	5.5	(7.8)	2.6	27.5	23.1	48.9	18.3	59.4	21.3	54.2	23.9	(174.1)	2.3
S3	0	1.8	7.8	1	33.3	14.6	52.8	20.4	62.4	29	52.9	26	(179.9)	1.3
S3*	0	9.7	7.5	1	24	14.6	35.9	16.5	42.3	25.1	51.5	26	(174.1)	1.3
S4	0	2.9	(8)	1.4	32.8	17.4	55.5	18.7	63.6	21.1	53.5	33.4	175.6	1.5
S4*	0	9.5	(7.8)	1.4	28.7	17.4	38.1	15.4	44.5	17.8	54.8	33.4	168.7	1.5

Note: Due to no organic phase is measured from the coupled analysis for specimens S2 and S4, and no pyrite phase is measured for specimens S2 and S3, the Young's moduli (values in the parentheses) are estimated by the averaged value measured from other specimens.

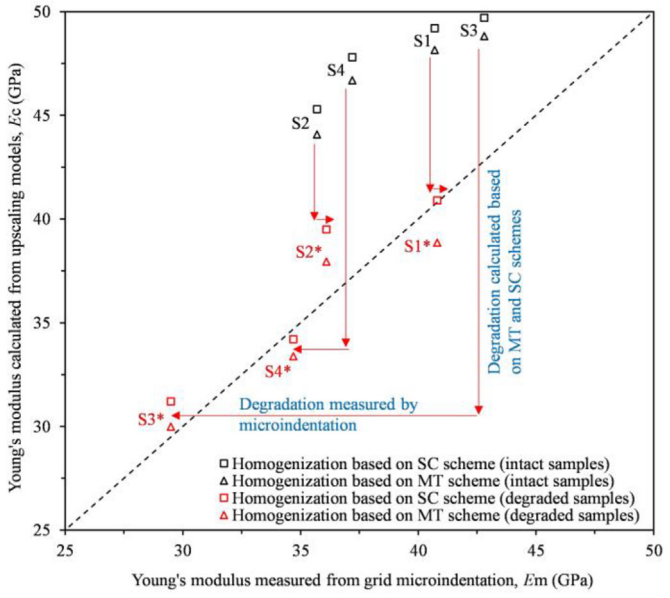


Fig. 11. The comparison between the calculated and the measured Young's modulus of intact and degraded shale composites.

Table 3
Results of the measured and calculated Young's modulus degradation of shale composites.

Sample	Measured and calculated degradation of Young's modulus (%)		
	Microindentation	MT	SC
S1	0.5	−19.3	−16.9
S2	1.1	−13.9	−12.8
S3	−31.2	−38.6	−37.2
S4	−6.5	−28.5	−28.3

4. Discussion

The relative contributions of clay and carbonate phases in mechanical degradation of shale composite are evaluated in this study. The carbonate dissolution triggers two effects on mechanical properties of the shale composite: increase in the areal voids and mechanical degradation of the carbonate phases. Assuming parameters of other constituent phases (both Young's moduli and area fractions) in the upscaling mechanical models remain the same, the carbonates dissolution at 20 °C, 0.1 MPa and at 80 °C, 8 MPa accounts for a reduction of the homogenized Young's modulus of Longmaxi shale composite by up to 10% and 33%, respectively. Clay minerals can swell after exposure to aqueous liquids (Madsen and Müller-Vonmoos, 1989), and consequently exhibit a reduction in their mechanical properties (Wong, 1998). Similarly, the degradation of Longmaxi shale composite attributed to clay swelling is estimated using the upscaling models. The resulted reduction in the homogenized E is up to 9% and 5% for exposure to DI water and 2% KCl solution, respectively. It is evident that, in the subsurface environment, the dissolution of carbonates under high temperature and pressure conditions dominates the mechanical softening of the Longmaxi shale formation. The XRD results indicate that the primary clay mineral in the studied samples is illite, which typically exhibits less swelling behavior compared to montmorillonite or chlorite (O'Brien and Chenevert, 1973). Therefore, for shale formations with more abundant

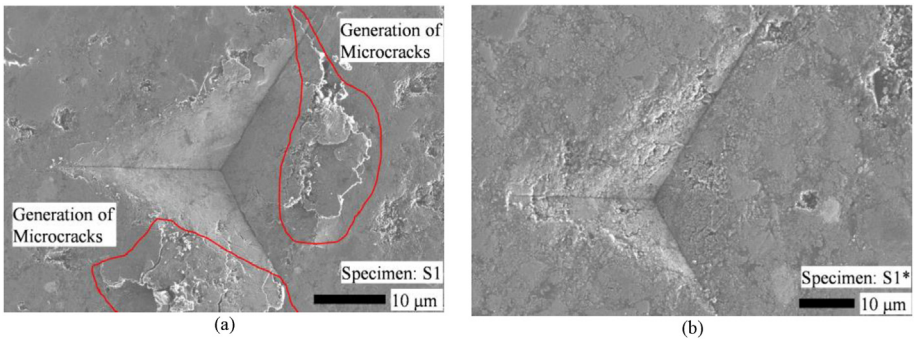


Fig. 12. Residual microindentation footprints of (a) intact and (b) degraded sample. Microcracks generated during the microindentation test are red circled.

montmorillonite or chlorite, swelling may play more significant role in the mechanical degradation after hydraulic fracturing.

Carbonate minerals may also act as inter-grain cementing agents in shale composites due to their precipitation within the marine sedimentary environment (Liu et al., 2021a). In this situation, the inter-grain bonding may be weakened by the dissolution of carbonates into hydraulic fracturing fluids. Therefore, research efforts are warranted to further study the roles of inter-grain cementation in mechanical degradation of shale formations. Moreover, though the short-term (7 d) softening behavior of the shale formation is investigated in this study, future works on mechanical degradation caused by long-term shale-fluid interactions are highly recommended.

In this study, based on the color change between the intact zone and the zone with water infiltration, the depth of shale-fluid interaction zone is hundreds of μm. Lu et al. (2020) reported that the degree of softening is not uniform within the interaction zone and the Young's modulus is approximately increased linearly with the increasing depth. Therefore, it can be estimated that along the infiltration depth, the Young's moduli of calcite, dolomite, and illite phases in the treated Longmaxi shale samples are degraded with a gradient up to 0.17, 0.20, and 0.09 GPa/μm, respectively. In future work, focused ion beam scanning electron microscopes (FIB-SEM) (Zhang et al., 2018) and computed tomography (CT) analysis (Chen et al., 2023) could be carried out to obtain high-fidelity 3D microstructural and mineralogical alterations induced by shale-fluids interactions.

5. Conclusions

This study presents a multiscale characterization of shale softening induced by water-based fluids at different temperatures and pressures. Mechanical alterations of shale constituent mineral phases are revealed using a coupled analysis. Multiscale mechanical models are established based on MT and SC schemes to assess the softening behavior of shale composite. Key conclusions obtained from this study are summarized as follows:

(1) After 7-d treatment of water-based fluids, E and H of clay phase are reduced by up to 28% and 30%, respectively and that of carbonate phases are reduced by up to 32% and 35%, respectively. Whereas, only minor mechanical alterations (less than 3%) are measured for organic, quartz, and pyrite phases. The KCl additive greatly mitigates mechanical degradation of the clay phase. The high temperature and pressure result in a mechanical degradation of carbonate phases as much as three times that occurs at 20 °C and 0.1 MPa.

(2) The upscaled softening behavior of shale composite is evaluated by the multiscale mechanical model. The degradation of clay and carbonate (i.e. calcite and dolomite) minerals results in a reduction of the homogenized E by up to 39%. Compared with the calculation results, smaller decreases in E were measured from the grid microindentation tests due to the massive microcracks generated in the intact shale samples.

(3) Carbonate dissolution dominates the mechanical softening of the Longmaxi shale samples, as it induces both the generation of voids and the mechanical degradation of carbonate phases. At 80 °C and 8 MPa, the carbonate dissolution results in a reduction in the E of shale composite by about 30%, while the degradation of clay minerals only causes a reduction by up to 9%.

Furthermore, in this study, the microstructure and mineralogy are assumed no significant changes along the depth. In future work, characterization of 3D microstructure of shale samples can be conducted to improve the evaluation of the multiscale softening behavior of shale.

CRediT authorship contribution statement

Jianting Du: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Andrew J. Whittle:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Liming Hu:** Writing – review & editing, Supervision, Funding acquisition. **Thibaut Divoux:** Writing – review & editing, Methodology, Investigation. **Jay N. Meegoda:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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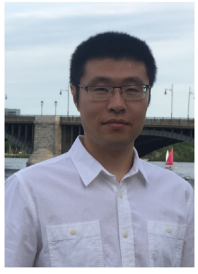
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